

Lab 7

1.

 $S \rightarrow OAO \mid 11B1 \mid BB$  $A \rightarrow C$  $B \rightarrow S \mid A$  $C \rightarrow S \mid \epsilon$ 

a.

 $N_0 = \{C\}$  $N_1 = \{C, A\}$  $N_2 = \{C, A, B\}$  $N_3 = \{C, A, B, S\}$  $N_4 = N_3$  $(C \rightarrow \epsilon)$  $(A \rightarrow C)$  $(B \rightarrow A)$  $(S \rightarrow BB)$  $\Rightarrow N = \{S, A, B, C\}$ 

Producción

Análisis

Nuevo

 $S \rightarrow OAO$ 

A

OAO, OO

 $S \rightarrow 11B1$ 

B

11B1, 11

 $S \rightarrow BB$ 

B, B

BB, B, B,  $\epsilon$  $A \rightarrow C$ 

C

C,  $\epsilon$  $B \rightarrow S$ 

S

S,  $\epsilon$  $B \rightarrow A$ 

A

A,  $\epsilon$  $C \rightarrow S$ 

S

S,  $\epsilon$  $C \rightarrow \epsilon$ 

-

 $\epsilon$  $S \rightarrow OAO \mid OO \mid 11B1 \mid 11 \mid BB \mid B$  $A \rightarrow C$  $B \rightarrow S \mid A$  $C \rightarrow S$

b.

 $S \rightarrow B, A \rightarrow C, B \rightarrow S, B \rightarrow A, C \rightarrow S$ 

Variables

|     |                                                                   |
|-----|-------------------------------------------------------------------|
| $S$ | $S \Rightarrow B \Rightarrow A \Rightarrow C = S: \{S, B, A, C\}$ |
| $A$ | $A \Rightarrow C \Rightarrow S \Rightarrow B: \{A, C, S, B\}$     |
| $B$ | $B \Rightarrow S, B \Rightarrow A \Rightarrow C: \{B, S, A, C\}$  |
| $C$ | $C \Rightarrow S \Rightarrow B \Rightarrow A: \{C, S, A\}$        |

 $S: 0A0 | 00 | 1B1 | 11 | BB$ 
 $A: \emptyset$ 
 $B: \emptyset$ 
 $C: \emptyset$ 
 $S \rightarrow 0A0 | 00 | 1B1 | 11 | BB$ 
 $A \rightarrow \emptyset \cdot A0 | 00 | 1B1 | 11 | BB$ 
 $B \rightarrow \emptyset A0 | \emptyset 0 | 1B1 | 11 | BB$ 
 $C \rightarrow \emptyset A0 | 00 | 1B1 | 11 | BB$ 

C.

 $P_0 = \{0, 1\}$ 
 $P_1 = \{0, 1, S, A, B, C\} \cdot (x \rightarrow \emptyset 00 \Rightarrow x \in \{S, A, B, C\})$ 
 $P_2 = P_1$ 

No Productor:  $\emptyset$ 

No alcanzables:

 $R_0 = \{S\}$ 
 $R_1 = \{S, 0, 1, A, B\} \cdot (S \rightarrow 0A0 | 1B1 | BB)$ 
 $R_2 = R_1$ 

(A, B solo generan 0, 1, A, B)

No derivable:  $\{C\}$

$S \rightarrow 0A0 \mid 00 \mid 1B \mid 11 \mid BB$   
 $A \rightarrow 0A0 \mid 00 \mid 1B \mid 11 \mid BB$   
 $B \rightarrow 0A0 \mid 00 \mid 1B \mid 11 \mid BB$

d.

$X_0 \rightarrow 00 \quad X_1 \rightarrow 01$   
 $S, A, B \rightarrow X_0 A X_0 \mid X_0 X_0 \mid X_1 B X_1 \mid X_1 X_1 \mid BB$

$X_0 A X_0 \Rightarrow X_0 D_1 \quad D_1 \rightarrow A X_0$   
 $X_1 B X_1 \Rightarrow X_1 D_2 \quad D_2 \rightarrow B X_1$

$S \rightarrow X_0 D_1 \mid X_0 X_0 \mid X_1 D_2 \mid X_1 X_1 \mid BB$   
 $A \rightarrow X_0 D_1 \mid X_0 X_0 \mid X_1 D_2 \mid X_1 X_1 \mid BB$   
 $B \rightarrow X_0 D_1 \mid X_0 X_0 \mid X_1 D_2 \mid X_1 X_1 \mid BB$   
 $D_1 \rightarrow A X_0$   
 $D_2 \rightarrow B X_1$   
 $X_0 \rightarrow 00$   
 $X_1 \rightarrow 01$

2.

$S \rightarrow aAa \mid bBb \mid \epsilon$   
 $A \rightarrow C \mid a$   
 $B \rightarrow C \mid b$   
 $C \rightarrow CDE \mid \epsilon$   
 $D \rightarrow A \mid B \mid ab$

g.

$N_0 = \{S, C\}$   
 $N_1 = \{S, C, A, B\}$   
 $N_2 = \{S, C, A, B, D\}$   
 $N_3 = N_2$

$(S \rightarrow \epsilon; C \rightarrow \epsilon)$   
 $(A \rightarrow C, B \rightarrow C)$   
 $(D \rightarrow A)$   
 $(C \rightarrow CDE; E \notin N)$   
 $\Rightarrow N = \{S, A, B, C, D\}$

$S \rightarrow aAa$   
 $S \rightarrow bBb$   
 $S \rightarrow \epsilon$

$A \rightarrow C$

$A \rightarrow a$

$B \rightarrow C$

$B \rightarrow b$

$C \rightarrow CDE$

$C \rightarrow \epsilon$

$D \rightarrow A$

$D \rightarrow B$

$D \rightarrow ab$

A

B

-

C

-

C

-

C, D

-

A

B

-

aAa, aa

bBb, bb

~~ε~~

C, ~~ε~~

a

C, ~~ε~~

b

CDE, DE, CE, E

~~ε~~

A, ~~ε~~

B, ~~ε~~

ab

$S \rightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow C \mid a$

$B \rightarrow C \mid b$

$C \rightarrow CDE \mid CE \mid DE \mid E$

$D \rightarrow A \mid B \mid ab$

b

$A \rightarrow C, B \rightarrow C, C \rightarrow E, D \rightarrow A, D \rightarrow B$

$S: \{S\}$   
 $A: A \Rightarrow C \Rightarrow E: \{A, C, E\}$   
 $B: B \Rightarrow C \Rightarrow E: \{B, C, E\}$   
 $C: C \Rightarrow E: \{C, E\}$   
 $D: D \Rightarrow A, D \Rightarrow B, D \Rightarrow X \Rightarrow E: \{D, A, B, C, E\}$   
 $E: \{E\}$

$S: aAa/aa | bBb/bb$

$A: a$

$B: b$

$C: CDE | CE | DE$

$D: ab$

$E: \emptyset$

$S \rightarrow aAa/aa | bBb/bb$

$A \rightarrow a/CDE | CE | DE$

$B \rightarrow b/CDE | CE | DE$

$C \rightarrow CDE | CE | DE$

$D \rightarrow ab/a/b/CDE | CE | DE$

C, No produccion:

$P_0 = \{a \cdot b\}$

$P_1 = \{a, b, S, A, B, D\} \cdot (S \rightarrow aAa, A \rightarrow a, B \rightarrow b, D \rightarrow ab)$

$P_2 = P_1$

(Todo  $id \in C$  contiene  $E \in P$ )

$S \rightarrow aAa/aa | bBb/bb$

$A \rightarrow a$

$B \rightarrow b$

$D \rightarrow ab/a/b$

b.

$$R_0 = \{S\}$$

$$R_1 = \{S, a, b, A, B\}$$

$$R_2 = R_1$$

$$\begin{aligned} (S \rightarrow aA \mid bB) \\ (A \rightarrow a, B \rightarrow b) \end{aligned}$$

B. no alcanzable

$$S \rightarrow aA \mid aA \mid bB \mid bB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

d.

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$$X_a \rightarrow a \quad X_b \rightarrow b$$

$$S \rightarrow X_a A X_a \mid X_a X_a \mid X_b B X_b \mid X_b X_b$$

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$$\begin{aligned} X_a A X_a &\Rightarrow X_a D_1 \quad D_1 \rightarrow A X_a \\ X_b B X_b &\Rightarrow X_b D_2 \quad D_2 \rightarrow B X_b \end{aligned}$$

CNF

$$S \rightarrow X_a D_1 \mid X_a X_a \mid X_b D_2 \mid X_b X_b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$D_1 \rightarrow A X_a$$

$$D_2 \rightarrow B X_b$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$